

Load Calculation for Auditorium						
Distribution Board						
SL no	Descripting of Items	Considered Rating(Watt)	Quantity	Total Wattage	D.F(%)	Maximum Demand(Watt)
i.	Lights					
	Circular LED Light	60	30	1800	60%	1080
	Flat LED Light	80	2	160	60%	96
	Wall bracket light (WL)	40	5	200	60%	120
	Ceiling light (CL)	40	2	80	60%	48
	13A -3pin (type-S)	1000	5	5000	60%	3000
	Wall Mounted Speaker	600	5	3000	100%	3000
	Total Maximum Demand(Including SDB1+SDB2+SDB3)					64314.1
	with 10% spare					70745.51
	Therefore, I					123.0269254
	with25% of RF					153.7836567
	Circuit Breaker : 160A TP MCCB					
	Cable: 4x1C-50mm2 NYY + 1x1c-16mm2 BYA as ECC					

Sub Distribution Board-01						
Serial #	Descripting of Items	Considered Rating(Watt)	Quantity	Total Wattage	D.F(%)	Maximum Demand(Watt)
i.	AC Loads					
	Central AC	3000	11	33000	100%	33000
	Total Maximum Demand					33000
	with 10% spare					36300
	Therefore, I					63.12594809
	with25% of RF					78.90743511
	Circuit Breaker : 80A DP MCB					
	Cable: 4x1C-25mm2 NYY + 1x1c-16mm2 BYA as ECC					

Sub Distribution Board-02						
Serial #	Descripting of Items	Considered Rating(Watt)	Quantity	Total Wattage	D.F(%)	Maximum Demand(Watt)
i.	Lights					
	Flat LED Light	80	4	320	60%	192
	Mirror Light	15	4	60	60%	36
	Tube Light	40	5	200	60%	120
	Wall bracket light (WL)	40	16	640	60%	384
	Stage Spot Light	800	3	2400	60%	1440
	Incandescent Rotating Stage Lights	1500	3	4500	60%	2700
	Ceiling light (CL)	40	8	320	60%	192
ii.	Fan					
	Ceiling Fan	75	3	225	50%	112.5
	Wall Mounted Fan	45	2	90	50%	45
iii.	Power Sockets					
	5A-2pin	300	6	1800	50%	900
	13A-3pin	1000	5	5000	60%	3000
iii.	AC Loads					
	AC(1.5 TON)	1500	2	3000	100%	3000
iv.	Others					
	Washroom Exhaust Fan	36	4	144	60%	86.4
	Total Maximum Demand					12207.9
	with 10% spare					13428.69
	Therefore, I					23.35258369
	with25% of RF					29.19072961
	Circuit Breaker : 32A DP MCB					
	Cable: 4x1C-4mm2 NYY + 1x1c-4mm2 BYA as ECC					

Sub Distribution Board-03						
Serial #	Descripting of Items	Considered Rating(Watt)	Quantity	Total Wattage	D.F(%)	Maximum Demand(Watt)
i.	Lights					
	Chandelier	400	1	400	100%	400
	Flat LED Light	80	4	320	100%	320
	Exhibition Spot Light	15	3	45	60%	27
	Mirror Light	15	10	150	60%	90
	Tube Light	40	8	320	60%	192
	Wall bracket light (WL)	40	2	80	60%	48
	Ceiling light (CL)	40	23	920	60%	552
	Door Light	30	2	60	100%	60
	Security Light	50	1	50	100%	50
	Garden Light	40	3	120	60%	72
ii	Power Sockets					
	5A-2pin	300	2	600	50%	300
	13A-3pin	1000	4	4000	50%	2000
iii.	Others					
	Washroom Exhaust Fan	36	7	252	60%	151.2
iv.	AC Loads					
	Central AC	3000	2	6000	100%	6000
	AC(1.5 TON)	1500	1	1500	100%	1500
	Total Maximum Demand					11762.2
	with 10% spare					12938.42
	Therefore, I					22.50000081
	with25% of RF					28.12500101
	Circuit Breaker : 32A DP MCB					
	Cable: 4x1C-4mm2 NYY + 1x1c-4mm2 BYA as ECC					